

CONTROLLED PRODUCT DOCUMENT
PRODUCT TECHNICAL SPECIFICATION
KAISTEEL Outdoor Kitchen System | Project-controlled technical baseline

Document Ref	KS-PTS-2026-V1.0
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Document Owner	KAISTEEL LIMITED (Hong Kong)
Product Family	Modular Outdoor Kitchen Systems
Status	Controlled Commercial Draft - Project Confirmation Required

CONTROL RULE This specification does not authorize manufacture by itself. Each order requires an approved model, bill of materials (BOM), shop drawing, utilities schedule and Technical Freeze Sheet. Any item marked TBC must be confirmed in writing before production release.

1. Purpose, Scope and Document Hierarchy

This document defines the technical boundary for KAISTEEL modular outdoor kitchen systems supplied for residential or commercial B2B projects. It is intended for quotation review, technical approval, manufacturing coordination, pre-shipment inspection, installation planning and warranty traceability.

Included	Primary structure, exterior and interior panel systems, cabinet modules, countertop and sink, canopy or shutter mechanism, configured electrical/lighting provisions, configured plumbing provisions, selected appliances, finish and export packaging.
Not automatically included	Site foundations, local structural engineering, utility connection, gas commissioning, destination permits, local installation, lifting/unloading, import clearance, certification testing, or third-party appliances unless expressly listed in the order.
Order of precedence	Signed sales contract / PI -> approved Technical Freeze Sheet -> approved shop drawings and BOM -> this specification -> catalog and other marketing materials.
Conflict rule	Where documents conflict, production shall stop and KAISTEEL shall issue a written clarification or revision. Marketing images and renderings are illustrative and do not override approved technical documents.

2. Product Identification and Approved Configuration

Customer / Distributor	_____
Project / Site	_____
PO / PI Number	_____
Model / Series	_____
Serial Number(s)	Assigned before final QC: _____
Use Classification	Residential [] Commercial []
Approved Drawing / BOM	Drawing Rev: _____ BOM Rev: _____ Freeze Date: _____

2.1 Commercial model reference

Model	Commercial Series	Nominal Envelope	Closure / Canopy	Control Status
KS-E2300	Essential	2300 x 880 x 2400 mm	Manual closure / canopy	Dimensions and final configuration subject to approved drawing
KS-P2300	Professional	2300 x 880 x 2400 mm	Electric actuator lift-up canopy	Actuator and control data TBC by project
KS-P2900	Premium	2900 x 880 x 2400 mm	Electric actuator lift-up canopy	Appliance and entertainment package by BOM
KS-R2900	Premium Enclosed	2900 x 960 x 2400 mm	Electric roller shutter	Shutter and interlock data TBC by project
KS-P2900-UP	Premium Upgraded	2900 x 960 x 2400 mm	Electric actuator lift-up canopy	Deep-worktop configuration by approved drawing

IMPORTANT Published weights, packed volumes and container quantities differ across current source materials. They are not controlled values in V1.0. Gross/net weight, packed dimensions and loading quantity shall be measured and recorded for each approved model before shipment.

3. System Construction and Materials

System	Controlled Requirement	Project / Manufacturer Confirmation
Primary frame	Q235 galvanized steel structural framework.	Section sizes, wall thickness, galvanizing grade/coating mass, weld standard and load basis: TBC.
Exterior panels	Aluminum exterior panel system suitable for the approved outdoor application.	Alloy, temper, thickness, finish system and fixing details: TBC.
Cabinet core / interior panels	Aluminum honeycomb panel system in configured cabinet/interior areas.	Face-sheet thickness, core thickness/density, adhesive/fire data and exact scope: TBC.
Water-resistant layers	SPC or other water-resistant layers only where shown on the approved BOM/drawing.	Material grade, thickness and location: TBC. No blanket claim applies.
Base / floor	Base construction and anchoring arrangement per approved project drawing.	Board/material type, thickness, live load, point load and anchoring: TBC.
Fasteners	Corrosion-resistant fasteners compatible with connected materials and outdoor exposure.	Material grade, coating and torque schedule: TBC.
Sealants / gaskets	Outdoor-compatible sealants and gaskets at designated joints and penetrations.	Brand/type, UV resistance, food-zone suitability and service temperature: TBC.

3.1 Worktop and sink

Worktop material	Solid SUS304 stainless steel. Material certificate / supplier declaration required for each production batch.
Nominal thickness options	6 mm / 8 mm / 10 mm. The selected thickness must be stated in the BOM and approved drawing.
Official texture names	Sand Time Floating Light Cracked Realm Mystic Engrave Infinity.
Texture process	TBC by manufacturer for each texture. Marketing names do not define the manufacturing process.
Edges / openings / upstands	As shown on the approved shop drawing. Edge profile, radius, cut-outs, drainage falls, rear/side upstands and tolerances must be dimensioned.
Sink	Factory-integrated or installed stainless steel sink as listed in the BOM. Stainless grade, gauge, dimensions, waste fitting, faucet and overflow: TBC by project.
Finish acceptance	Uniform approved texture and direction; no structural cracks, sharp edges, unacceptable dents, deep scratches, contamination or visible repair outside the approved sample standard.

3.2 Cabinetry, storage and hardware

Cabinet modules	Aluminum / aluminum-honeycomb cabinet construction as listed in the approved BOM.
Doors and drawers	Free movement, controlled gaps, positive latching and no interference through full travel.
Hardware	Outdoor-suitable hinges, slides, handles and locking parts; make, grade, load rating and corrosion class: TBC.
Shelving	Integrated organizational shelving only where listed in the BOM; load rating: TBC.
Color	Standard commercial references: black, white/gray or silver variants. Exact color code, gloss and approved sample govern production.

4. Canopy, Shutter and Mechanical Systems

Item	Requirement	Acceptance / Data Required
Manual mechanism	Smooth controlled movement and secure retention at intended positions.	Operating force, retention method, cycle test and safety provision: TBC.
Electric lift canopy	Electric actuator system matched to load and travel; controls located and protected as approved.	Actuator make/model, voltage, stroke, load, duty cycle, IP rating, limit protection and wiring: TBC.
Electric roller shutter	Complete opening/closing without binding; secure in closed state.	Motor, rated load, controls, limit settings, manual override, interlock and IP rating: TBC.
Hinges / pivots	Aligned, securely fixed and free from abnormal play or noise.	Material, fastener grade, lubrication and torque: TBC.
Pinch / crush risk	Moving systems shall include project-approved safeguards, warnings and operating instructions.	Risk assessment and destination requirements: project responsibility / TBC.
Functional cycle test	Minimum factory cycle test before shipment.	Cycle quantity: TBC; record actual test in QC checklist.

5. Electrical and Lighting

NO IMPLIED CERTIFICATION Electrical components and assemblies are project-specific. CE, UKCA, UL, ETL, CSA, RCM or other certification shall not be claimed unless a valid certificate and exact model/scope are recorded in the order file.

Supply	Voltage, frequency, phase, plug/connection, rated input and maximum current: TBC for destination and project.
Distribution / protection	Outdoor distribution box and protective devices only as stated in BOM. RCD/GFCI/RCBO rating, breaker schedule and enclosure IP rating: TBC.
Earthing / bonding	Protective earthing and equipotential bonding points to be provided where required by the approved electrical design and destination rules.
Wiring	Cable type, conductor size, temperature rating, routing, strain relief, terminals, glands and separation from heat/water zones: TBC.
Lighting	Recessed and/or LED ambient lighting as configured. Color temperature, power, driver, dimming and IP rating: TBC.
Sockets / track power	Quantity, type, rating and destination compatibility per BOM. Covers and outdoor suitability shall be confirmed.
Factory test	Visual wiring inspection, continuity, protective-earth continuity, insulation resistance and functional checks as applicable. Test limits and instrument details shall be recorded.
Site connection	Final connection, inspection and certification by a qualified person under destination requirements unless expressly included in KAISTEEL scope.

6. Plumbing, Drainage and Gas Interfaces

Interface	Factory Scope	Project Confirmation / Site Responsibility
Cold / hot water	Internal pipework and prepared connection only where listed in BOM.	Pipe material/size, pressure rating, inlet size/location, hot-water provision and pressure test: TBC.
Waste	Sink waste and internal drainage arrangement where configured.	Outlet size/location, trap, fall, venting and local connection: TBC.
Leak testing	Factory leak test where internal plumbing is supplied.	Test medium, pressure/duration and acceptance limit: TBC; record in QC.
Gas appliance	Appliance and flexible/internal connection only if expressly listed and approved.	Fuel type, inlet pressure, regulator, hose/pipe standard, shut-off, ventilation and certification: TBC.
Gas commissioning	Not included unless expressly agreed.	Must be completed by a qualified local gas installer in accordance with destination requirements.

7. Appliances and Optional Equipment

Configuration basis	All grills, cooktops, range hoods, refrigerators, displays/TVs, speakers and other equipment must be identified by manufacturer, model and quantity in the BOM.
Typical commercial options	4- or 6-burner gas/charcoal grill, induction cooktop, range hood, refrigerator, entertainment display and outdoor audio. Typical descriptions are not a commitment of supply.
Compatibility	Dimensions, ventilation clearances, heat shielding, electrical/gas requirements, service access and destination suitability shall be checked before Technical Freeze.
Certificates	Copy exact manufacturer declarations/certificates into the order file where required. Certificate scope must match the supplied model.
Warranty	Third-party appliances are covered solely by the appliance manufacturer's warranty unless the signed contract expressly states otherwise.

8. Environmental and Installation Conditions

Condition	Requirement / Limitation
Outdoor exposure	Product is designed for configured outdoor use but is not represented as permanently watertight or flood-proof. Close the canopy/shutter and follow the maintenance guide when not in use.
Coastal / saline sites	Enhanced cleaning and corrosion-control measures are required. Material/finish selection and maintenance interval must be project-reviewed.
Wind	Site wind load, anchoring, closed/open operating limits and local structural compliance are not established by this generic specification; project engineering is required.
Temperature / snow / seismic	Design values and suitability are TBC by project and destination.
Foundation	Level, stable, adequately drained and capable of supporting verified equipment and imposed loads. Tolerances and anchor layout per approved drawing.
Clearances	Maintain appliance manufacturer clearances, ventilation and service access. Do not obstruct drainage or electrical ventilation.
Installation	Assembly, lifting, anchoring and utility connection by qualified personnel using approved instructions and local requirements.

9. Surface Finish, Workmanship and Dimensional Control

Characteristic	Default Acceptance Principle	Project-specific Value
Overall dimensions	Measured against approved drawing; no interference with installation or operation.	Tolerance: TBC
Level / squareness	Stable, square and visually aligned; doors, drawers and moving systems operate freely.	Tolerance: TBC
Panel gaps / flushness	Consistent with approved sample and drawing.	Tolerance: TBC
Welds	Complete and structurally sound; no visible cracks, burn-through, sharp spatter or unapproved repair.	Weld process / acceptance standard: TBC
Coating / panel finish	Uniform approved color/gloss; free from peeling, blistering, exposed substrate and unacceptable dents/scratches.	Color code, gloss and DFT: TBC
Stainless texture	Correct official texture, uniform orientation and approved appearance.	Approved sample / finish ID: TBC
Sharp edges	No accessible hazardous sharp edges or burrs after normal assembly and use.	Mandatory

10. Factory Inspection, Traceability and Release

Traceability	Each completed unit shall carry a unique Serial Number linked to PO/PI, model, drawing/BOM revisions, material records, inspection date, inspector and photo record.
Inspection basis	The current KAISTEEL Pre-Shipment Quality Inspection Checklist V1.0, approved project documents and applicable supplier test records.
Material evidence	Retain SUS304 evidence, structural material declarations and finish/batch data reasonably available for the order.
Nonconformity	Safety, structural, functional, identity or missing-part failures place the unit on HOLD. Rework requires documented reinspection.
Final release	Only an authorized KAISTEEL representative may release the unit after inspection status is PASS and required evidence is complete.

11. Packaging, Marking and Shipping Data

Item	Controlled Requirement
Protection	Clean and dry product; protect stainless surfaces, corners, projecting parts, moving components, glass/screens and appliances against handling and moisture.
Restraint	Lock or brace doors, drawers, canopy/shutter and loose components. Prevent movement inside the package.
Packaging	Standard export protection plus crate/frame where ordered. Wood packaging must meet destination requirements, including ISPM 15 where applicable.
Marking	PO/PI, model, Serial Number, case number, gross/net weight, dimensions, center of gravity/handling marks and destination marks as required.
Measured shipping data	Record actual packed L x W x H, gross/net weight and package count. Do not rely on historical catalog estimates.
Document pack	Packing list, commercial invoice, approved QC release, photo record, installation/operation documents, warranty information and applicable certificates.

12. Warranty Alignment

Primary structure	Residential: 15-Year Limited Commercial: 15-Year Limited. Covers solid SUS304 worktop and structural steel frame defects in material and workmanship, subject to the Warranty Certificate.
Surface finish	Residential: 5-Year Limited Commercial: 5-Year Limited. Natural weathering, fading and surface oxidation are excluded as stated in the Warranty Certificate.
Mechanical components	1 Year for residential and commercial use, including configured actuators, canopy motors and LED lighting systems.
Third-party appliances	Manufacturer Warranty only.
Governing document	KAISTEEL Limited Warranty Certificate V1.4 or the later certificate expressly incorporated into the signed order. This specification does not expand warranty coverage.

13. Project Confirmation Schedule

Required Before Production Release	Value / Reference	Status / Approval
Final model, dimensions and layout	_____	Open ☐ Approved ☐
Drawing and BOM revision	_____	Open ☐ Approved ☐
Frame sections, thickness and corrosion system	_____	Open ☐ Approved ☐
Countertop thickness, texture and edge details	_____	Open ☐ Approved ☐
Panel materials, thickness and finish code	_____	Open ☐ Approved ☐
Canopy/shutter mechanism and controls	_____	Open ☐ Approved ☐
Electrical schedule and destination compatibility	_____	Open ☐ Approved ☐
Water, drainage and gas interfaces	_____	Open ☐ Approved ☐
Appliance make/model/certification	_____	Open ☐ Approved ☐
Packaging, measured shipping data and marks	_____	Open ☐ Approved ☐

14. Approval and Change Control

VERSION CONTROL A signed approval does not authorize unrecorded substitutions. Any change after Technical Freeze requires a written Engineering Change Order describing the technical, price and delivery effect.

Role	Name / Company	Signature	Date
Prepared by	KAISTEEL	_____	_____
Technical review	_____	_____	_____
Manufacturer confirmation	_____	_____	_____
Customer / Distributor approval	_____	_____	_____

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